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The Brillouin linewidth in water as a function of temperature and salinity: the missing empirical relationship

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Abstract

Brillouin scattering in water has been proposed as a method to measure temperature profiles in the ocean. Spontaneous Brillouin scattering can be understood as light scattering on phonons in the water originating from thermal fluctuations. The scattering signal usually features three peaks in its spectrum, of which one is attributed mainly to scattering from suspended particles and shows no frequency shift from the incident light, while the other two are shifted by the phonon frequency leading to a linear dependency of this Brillouin shift to the speed of sound. The speed of sound is dependent on temperature and salinity. To measure temperature and salinity independently, a second measurement parameter is necessary. The spectral width of the Brillouin lines constitutes a measure for the phonon lifetime and depends on various macroscopic water properties. This linewidth was investigated in a laboratory using a scanning FPI (Fabry-Perot interferometer) over a temperature range from 1.8 °C to 35 °C and 13 salinities from 0 ppt to 36 ppt. A strong temperature dependence was confirmed and a weaker, nonlinear dependency on salinity was found. An empirical model was created to describe the data for the purpose of enabling remote sensing of water temperature profile and salinity simultaneously in the future.

1 Introduction

The possibility of remotely sensing depth-resolved temperatures in the ocean by using a Brillouin LIDAR (light detection and ranging) was first proposed by Hickman et al. [1] and demonstrated for the first time under laboratory conditions by Rudolf et al. [2]. The physical quantity exploited in this experiment is the Brillouin shift of the backscattered signal, which constitutes a direct measurement of the speed of sound in water. The basic principles of Brillouin scattering were studied in [3] and [4]. The Brillouin shift ν_B is directly proportional to the refractive index n and the speed of sound v_S , which are both functions of the temperature T and salinity S . Both, the index of refraction and the speed of sound have an additional dependency on pressure, which are here neglected due to simplicity, but must be taken into account in a real experiment in the ocean. For a fixed incident wavelength λ and scattering angle θ the Brillouin shift is then represented by

$$\nu_B(T, S) = \pm \frac{2n(T, S)}{\lambda} v_S(T, S) \sin(\theta/2), \quad (1)$$

with the scattering angle usually being set to 180 ° corresponding to measuring the direct backscattered light. Of the five parameters ν_B , v_S , n , T and S , only ν_B is measured. For the index of refraction and speed of sound empirical relations as a function of temperature and salinity exist [5–7]. They are

$$n(T, S) = n_0 + S(n_1 + n_2T + n_3T^2) + n_4T^2 + \frac{n_5 + n_6S + n_7T}{\lambda} + \frac{n_8}{\lambda^2} + \frac{n_9}{\lambda^3} \quad (2)$$

for the index of refraction [5] and

$$v_S(T, S) = c_0 + c_1T + c_2T^2 + c_3T^3 + c_4S + c_5S^2 + c_6ST + c_7ST^2 \quad (3)$$

for the speed of sound [6] with the coefficients given in Tables 1 and 2, respectively.

Along with Eq. (1) no well-defined problem is found as one additional parameter has to be determined. As salinity has varied only marginally since 1900 for a fixed location and time of year, salinity has been chosen as a known quantity so far. The validity of this approach has already been

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Table 1 Coefficients for the empirical relation Eq. (2) describing the index of refraction of sea water as a function of temperature T and salinity S

n_0	1.31405	n_1	$1.779 \cdot 10^{-4}$
n_2	$-1.05 \cdot 10^{-6} \text{ }^\circ\text{C}^{-1}$	n_3	$1.6 \cdot 10^{-8} \text{ }^\circ\text{C}^{-2}$
n_4	$-2.02 \cdot 10^{-6} \text{ }^\circ\text{C}^{-2}$	n_5	15.868 nm
n_6	0.01155 nm	n_7	$-0.00423 \text{ nm }^\circ\text{C}^{-1}$
n_8	-4382 nm^2	n_9	$1.1455 \cdot 10^6 \text{ nm}^3$

Table 2 Coefficients for the empirical relation Eq. (3) describing the speed of sound in the sea water as a function of temperature T and salinity S

c_0	1402.392 ms^{-1}	c_1	$5.01109398873 \text{ ms}^{-1}^\circ\text{C}^{-1}$
c_2	$-0.0550946843172 \text{ ms}^{-1}^\circ\text{C}^{-2}$	c_3	$0.00022153596924 \text{ ms}^{-1}^\circ\text{C}^{-3}$
c_4	$1.32952290781 \text{ ms}^{-1}$	c_5	$0.000128955756844 \text{ ms}^{-1}$
c_6	$-0.012562783426 \text{ ms}^{-1}^\circ\text{C}^{-1}$	c_7	$0.000096840315641 \text{ ms}^{-1}^\circ\text{C}^{-2}$

shown by Fry et al. in terms of accuracy limitations [8]. A simultaneous measurement of both temperature and salinity is still very much appropriate, especially as it is unclear whether the approach mentioned above would hold true in the future in the light of global climate changes and melting ice caps at the poles. Additionally, expanding this remote sensing application with a direct salinity measurement would bring it on par with already established, contact-based methods, such as CTD (*conductivity, temperature, depth*) probes. To circumvent this reference to historical salinity data, one additional relation or measured quantity would be required. A promising candidate for this purpose is the spectral width of the Brillouin scattering, which also depends on both temperature and salinity and is given by

$$\Delta\nu_B = \frac{\Gamma(T, S)}{2} \left(\frac{4\pi n(T, S)}{\lambda} \sin(\theta/2) \right)^2, \quad (4)$$

with the incident wavelength λ , the index of refraction n and the damping Γ , which in turn is defined by the relation

$$\Gamma = \frac{1}{\rho} \left(\frac{4}{3}\eta_s + \eta_b + \frac{\kappa}{C_p}(\gamma - 1) \right). \quad (5)$$

It contains the density of water ρ , the shear viscosity η_s , the bulk viscosity η_b , the thermal conductivity κ , the specific heat at constant pressure C_p and the specific heat ratio $\gamma = C_p/C_v$ with the specific heat at constant volume C_v . These quantities are in turn dependent on either T or S or both, making it difficult to achieve simulations. Also, the damping cannot be assumed to be independent of frequency, which means there is little to no available data for a robust simulation. Measurements of bulk and shear viscosity have

been performed successfully at optical frequencies, but only for changes in temperature [9, 10]. With these considerations in mind, the most promising approach is finding an empirical relation between the Brillouin linewidth, temperature and salinity. Attempts of measuring the spectral width of the Brillouin scattering in water have been made in the past by Fry et al. [11]. However, the density of the data was still insufficient for generating an empirical model. While the methodology of this work is similar to that presented in [11], there are subtle differences between the experiments. Both setups use a Fabry-Pérot interferometer for the frequency measurement. While the one in [11] is designed as a confocal FPI, the one used in the work presented here is a plane-plane FPI with a higher free spectral range of 23 GHz. In both cases, a spatial filter is used prior to coupling the signal into the FPI. Also, the laser used in [11] was a frequency-doubled injection-seeded Nd:YAG laser with a much higher bandwidth of approximately 100 MHz. Thus, the spectra were taken by slowly scanning the etalon putting much higher constraints on mechanical stability. The laser system employed in our setup operates in cw-mode and thus features a significantly narrower linewidth, which allows to approximate the elastic line as the FPI response function without deconvolution with the laser line. The probe chamber in [11] was a horn-shaped water cell that was mounted vertically to the optical table to reduce back reflections. However, temperature control was not as accurate as temperature changes were achieved by heating wires. In our experiment, changes in temperature were performed by changing a constant temperature bath. The goal of this work is to achieve a high density of linewidth measurements in both temperatures and salinities and generate an empirical model from the data, enabling a simultaneous measurement of temperature and salinity through Brillouin scattering.

2 Experimental setup

The experimental setup is shown schematically in Fig. 1. An ECDL (*external cavity diode laser*) based fiber amplifier in MOPA (*master oscillator power amplifier*) configuration is used as a light source, generating up to 6 W at 1060 nm. The light is then frequency doubled by resonant enhancement to achieve output powers of up to 3 W at 530 nm, with the laser being operated at 2.5 W for all experiments. The laser radiation then reaches the probe chamber, which is a custom-built, double-walled cylinder made of stainless steel with a length of 50 cm and a probe chamber volume of 600 ml with wedged, anti-reflection coated windows on either side. Both windows are also flushed by compressed air to suppress condensation at lower temperatures. The inner chamber is used for the probe water, while the outer ring contains the water from the heating/cooling cycle. With this approach, the

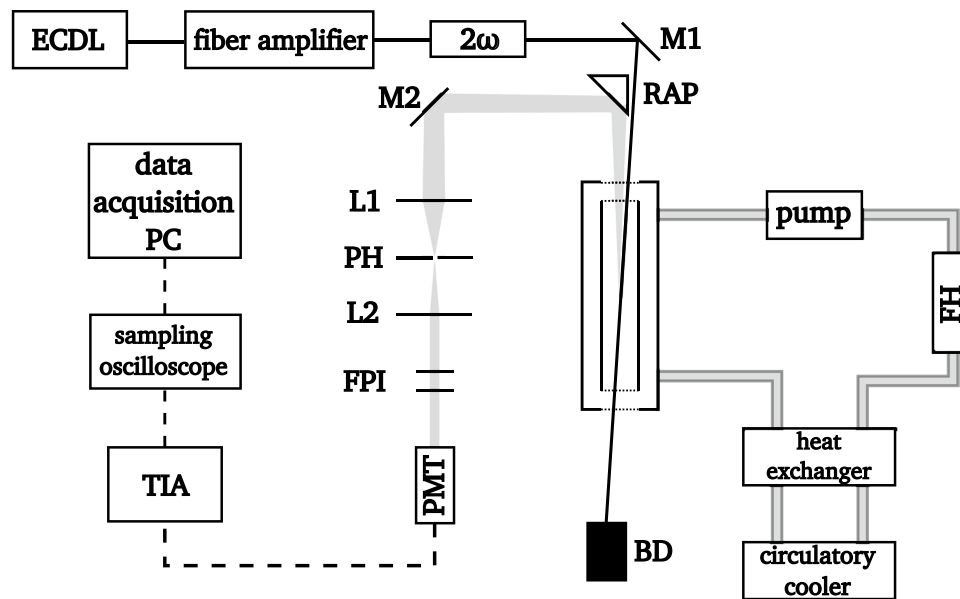


Fig. 1 Schematic view of the experimental setup. Light at 530 nm generates spontaneous Brillouin scattering in a temperature-controlled water cell. The backscattered light is collected at a scattering angle of 179° by a right-angle prism (RAP) and sorted from unwanted light sources in a spatial filter. The signal is analyzed in a scanning Fabry-Perot etalon (FPI) and detected by a photomultiplier tube

(PMT). The electrical signal passes through a transimpedance amplifier (TIA) and is analyzed by a sampling oscilloscope that transmits the collected spectra to a lab PC via network connection. Remaining abbreviations are external cavity diode laser (ECDL), mirrors M1 and M2, lenses L1 and L2, pinhole PH, flow heater (FH), beam dump (BD)

probe water does not require circulation, which greatly benefits maintaining a good ratio between elastic and Brillouin scattering throughout a measurement as the exposition of suspended particles to the laser beam is minimized. Heating and cooling are split into two separate cycles, connected by a heat exchanger. Heating is achieved by a flow heater going up to 309 K, while cooling is enabled by a circulatory cooling unit (DLK 402 DP, Fryka Kältetechnik GmbH) achieving a minimal temperature of 269 K. By keeping these cycles separate, a mixing of the cooling liquid and the water reaching the probe chamber is made impossible. The temperature is monitored by a digital thermometer with an accuracy of ± 0.1 K. All data in this work was collected from high to low temperatures for the sake of consistency, while comparisons with measurements recorded from low to high temperatures showed only minute influences from hysteresis.

The backscattered light is collected at $179^\circ \pm 0.5^\circ$ by a right-angle prism with an anti-reflection coating and guided to a spatial filter consisting of a circular pinhole with a $30\text{ }\mu\text{m}$ diameter and two lenses L1 and L2 with focal length $f_1 = 75\text{ mm}$ and $f_2 = 50\text{ mm}$, respectively. The spatial filter is essential for the experiment to guarantee only small angle deviations that would otherwise artificially broaden the spectrum according to Eq. (4) and to exclude elastic scattering originating from reflections off the various glass surfaces of the experiment. The spatial filter also makes it possible to obtain light only from a small fraction of the water pillar

aligned to the center of the filter. Moreover, errors from temperature gradients and delays are minimized by placing the thermometer close to the water volume probed.

The spatially filtered light reaches a home-built plane-plane scanning FPI with a free spectral range (FSR) of approximately 23 GHz chosen to avoid overlapping of neighboring Stokes and Anti-Stokes lines over the entire achievable temperature range. It reaches a finesse of up to 280 for laser light and typically 50 to 60 for scattered light. It uses two plane-plane mirrors with a reflectivity of 0.996 and a surface quality of $\lambda/10$ at 546.1 nm (coating by Layertec GmbH). It is installed vertically to the side of the optical table to avoid influences of gravity on the scan. The lower mirror is scanned by three piezoelectric transducers, controlled by a LabVIEW program with separate voltage ramps for each transducer. The output voltages are amplified by a 3-axis piezo controller (Thorlabs MDT639B) before reaching the actuators. Each scan covers about seven FSR with a 2.5 Hz scan rate. The transmitted light is then detected by a photomultiplier tube (Hamamatsu R6358). The generated photocurrent is amplified by a transimpedance amplifier (Analog Devices EVAL-ADA4625-1ARDZ) before reaching a sampling oscilloscope (Tektronix TDS5034B), transmitting the spectra to the acquisition PC via network.

Pure water of a particular salinity is prepared at a time, brought to a certain temperature and then scanned from high to low in the range from 308 K to 271.4 K while

spectra are collected continuously. The base for every probe was highly demineralized and UV-treated water (Milli-Q, Merck), mixed with sea salt (Tropic Marin Pro Reef) containing all minerals of sea water in their natural proportions. This provides a consistent way of obtaining water probes at very specific salinities. To avoid precipitation, each probe has to be prepared freshly before the experiment while still ensuring that the added salt is fully dissolved. When mixing a probe at a certain salinity a precision scale was used to weigh the added salt. The amount added was calculated from

$$m_{\text{salt}} = \frac{VS}{1000 - S} \quad (6)$$

with the salt mass m_{salt} in grams, the Salinity S in ppt and the probe volume V in ml which was set to 800 ml for all mixed probes. Equation 6 also accounts for the salt's own mass in the probe. When changing probes, the probe chamber was always flushed with pure water twice if advancing from lower to higher salinities to remove any remains of salt from the previous measurement. If going from a higher to a lower salinity, the chamber was flushed at least three times instead with the amount depending on the distance between salinities.

With the described procedure a wide range of salinities were probed.

3 Results

An example of an obtained spectrum and a zoomed-in triplet at 0 ppt and 307.5 K is shown in Fig. 2 along with a Lorentzian fit. The fit excellently describes both the elastic and inelastic peaks, allowing for an approximation that greatly simplifies deconvolution later on. The actual data analysis is performed in Python by the following procedure: All spectra collected at a certain salinity are first sorted by temperature. For a single data point a maximum of 25 spectra are used for evaluation, if more exist for one point of temperature and salinity a second, independent evaluation is performed. The first of these 25 spectra is taken as a reference and the remaining 24 spectra are then shifted relatively to the first to align all peaks. This compensates for drifts and fluctuations of the FPI during data acquisition of the set of data. Since the piezos of the etalon might not behave linearly, the shifted spectra are averaged and linearized by searching for peak positions in the averaged spectrum and fitting a polynomial of order three to these positions. From this polynomial, the spectra are corrected to represent a linear time axis. Lorentzian functions are then fitted at every peak as well as at every Stokes and anti-Stokes line to obtain a width in the time domain. To convert the time axis to a frequency axis the well-known relation between the Brillouin shift, temperature and salinity (Eq. (1) along with relations from [6] and [5]) are utilized. Using the known frequency shift for a

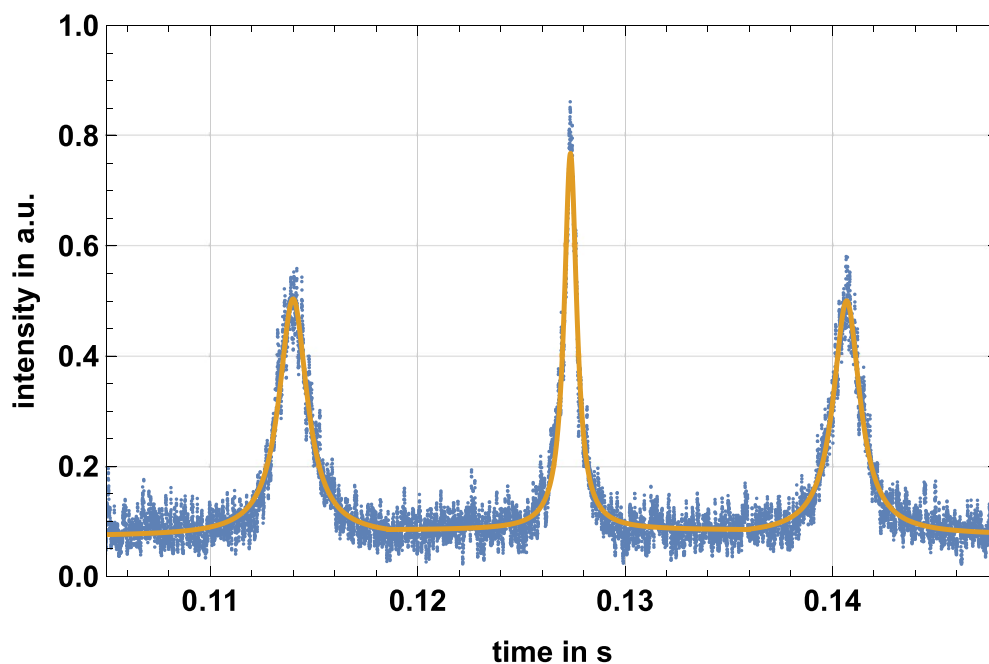


Fig. 2 A triplet as it occurs in all spectra, extracted at 0 ppt and 307.5 K. It is easy to identify the elastic scattering line (center) as well as the Stokes (left) and anti-Stokes line (right) from Brillouin

scattering. A Lorentzian line is fitted to each of these peaks. It can be seen that a Lorentzian lineshape makes for an excellent approximation of those peaks

given temperature-salinity value, we can calibrate the time axis of the etalon scan to a corresponding frequency axis and thus the width of the shifted lines can be determined. The width obtained through this procedure is still convolved with the response function of the FPI (quantifying the FPI's resolution. Since the laser linewidth is much smaller than the FPI's resolution, the width of the response function can be approximated by the linewidth of the elastic peak. By taking advantage of the above Lorentzian approximation, the true Brillouin width is obtained by simply taking the difference of the elastic and inelastic width [12]. With this, the width of every Stokes and anti-Stokes line in the spectrum is obtained and averaged. The obtained values are also filtered by various criteria, most importantly if the calculated FSR lies too far from the expected value ($> \pm 1$ GHz), only one triplet was fitted or the relative error in the Brillouin width being too large ($> 20\%$).

4 Discussion and conclusion

An abridged collection of temperature curves is shown in Fig. 3 for the sake of visibility, along with a fourth order polynomial fitted to each curve. A sharp drop in the linewidth from 0ppt to 5ppt is observed. The nonlinear temperature behavior as first reported in [11] is clearly reproduced. The data of Fry et al. also described a plateau near the point of highest density at 276.95K that could not be reproduced in a similar experiment by one of that paper's co-authors in Beijing (see also [11]). In the data presented here, this plateau is visible in some, but not all temperature curves, for yet unexplained reasons. Furthermore, the increase in linewidth at temperatures below 288 K is less extreme than in the previous works mentioned above. Possible explanations for this

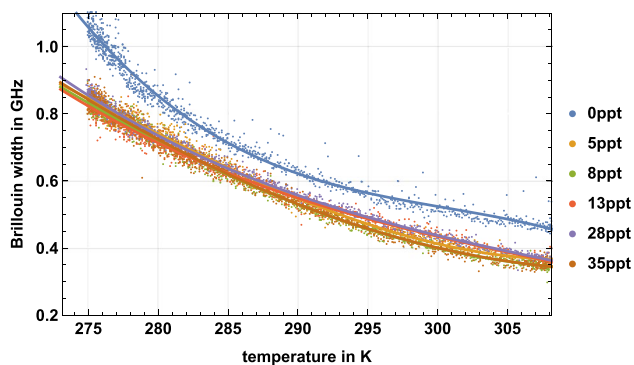


Fig. 3 Measured temperature dependency curves for selected values of the salinity with fourth-order polynomial fits. A sharp drop in Brillouin width as soon as salt is introduced to the probe as well as a nonlinear behaviour can be seen. Salinities that were also measured, but are not shown in this figure for the sake of visibility are 3ppt, 15ppt, 20ppt, 32ppt, 34ppt, 36ppt

different behavior might be related to the dew point temperature and a distortion of the signal by condensation, though a full explanation is not inherently clear as the exact laboratory conditions in these other experiments are unknown to us. Our data provides the highest density in both temperature and salinity available to the best of our knowledge and will now be described by an empirical model, similar to already existing empirical relations in [6] and [5].

Polynomials of order three were fitted to the data yet none of these was capable of describing the data accurately, so all possible fourth-order polynomials were fitted to the data instead and rated by their fit residuals and relative errors. When the error of a parameter exceeded the value itself, this polynomial contribution was eliminated from the fitting polynomial and the procedure was repeated. The best-obtained fit is shown in Fig. 4 along with all data points. The polynomial itself is shown in Table 3. The uncertainty for the temperature measurement ΔT is ± 0.1 K for all samples and temperatures and is determined by the accuracy of the digital thermometer. For salinity, the uncertainty ΔS can be determined by using Gaussian error propagation on Eq. (6). It rises linearly from ± 0.029 ppt at 3 ppt up to ± 0.3 ppt at 36 ppt. The measured salinities were chosen by the frequency of their natural occurrence. An especially dense probing was made around 35 ppt, with this being the ocean average value. A higher density at lower salinity made a reasonable choice due to fresh water basins or low salinity seas such as the Baltic sea with a salinity of 7 ppt [13]. The full data once again confirms only a small dependency on salinity with the clearly visible temperature dependency, especially below 288 K. The influence of salinity on the spectral width appears to be largely nonlinear, with the largest influence leaning towards lower salinities. The drop from 0 ppt up until 13 ppt is most noticeable and reflects observations that can be made from [11]. At higher salinity, the effect appears to saturate, except for a slight bump at 20 ppt.

The fit quality is further checked by calculating the deviation of each data point from the fit curve. Figure 5 shows the absolute deviation from the fit in a heat map style in MHz. White spaces indicate spots where no data of sufficient quality was available. More than 50% of data points have a deviation of less than 10 MHz or 2%, more than 97% deviate less than 40 MHz or 6%, marking an overall excellent data description by this polynomial.

This empirical relation of the spectral width of the Brillouin scattered light in water as a function of temperature and salinity is of great use for LIDAR remote sensing applications towards a simultaneous determination of temperature and salinity. For this particular application, the relatively weak dependency of the spectral width on salinity as observed in this work turns out to be advantageous. Especially, at higher salinities typical for the ocean, any change in the spectral width can be largely attributed to temperature

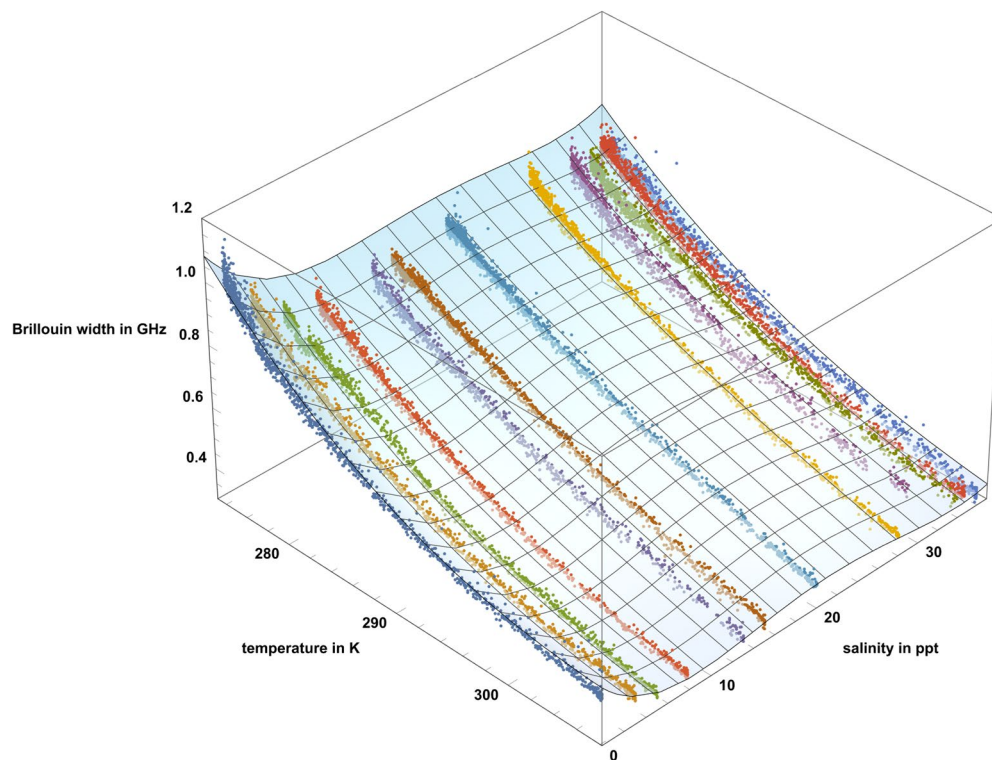


Fig. 4 All measured Brillouin widths depending on both temperature and salinity along with an empirical polynomial of fourth order fitted to all temperature curves. The fit plane is partly transparent, data

points below the curve appear lighter in color. A strong dependence on temperature is clearly visible while the effect of salinity on the linewidth appears relatively small by comparison.

Table 3 Parameters of the optimal fit polynomial for the data presented in Fig. 4 along with their standard error

Parameter	Estimate	Standard error
1	137.979	6.926
S	-3.2202	0.0659
T	-1.2322	0.0715
S^2	0.07899	0.00194
T^2	0.003656	0.000246
ST	0.02167	0.000454
S^3	-1.474×10^{-4}	1.589×10^{-6}
T^3	-3.586×10^{-6}	0.2818×10^{-6}
S^2T	5.105×10^{-4}	1.334×10^{-5}
ST^2	-3.685×10^{-5}	0.07829×10^{-5}
S^4	1.7328×10^{-6}	0.02396×10^{-6}
S^2T^2	8.6874×10^{-7}	0.22992×10^{-7}

Salinity S is to be provided in ppt, while temperature T has to be entered in Kelvin to obtain a Brillouin width in GHz. Relative errors are between 1.1% and 7.9% across all parameters

changes. Thus, due to the strong dependence of the spectral shift on both temperature and salinity over the full range from 273.15K to 308.15K and 0ppt to 36ppt, respectively, the temperature is best determined through the measurement of

the spectral width. With the temperature known, the salinity can then be measured by utilizing the spectral shift at salinities greater than 8ppt and temperatures below 288.15K. For higher temperatures and especially salinities below 8ppt, reversing that order might be advantageous. Prior to a measurement only very basic assumptions about the expected regime of temperature and salinity would be needed to make this approach feasible.

Measurements of the temperature profile by Brillouin scattering can be performed by employing our pulsed laser system detailed in ref. [2]. It operates at 543.3 nm and is based on an Rubidium edge-filter (Excited State Faraday Anomalous Optical Filter ESFADOF). One ESFADOF can measure the Brillouin shift as was demonstrated in [2]. Expanding this setup by a second, slightly detuned ESFADOF allows measuring the Brillouin width simultaneously.

In summary, the findings of this work only cement the proposition that the spectral width makes for a useful quantity in resolving the discussion of the introduction regarding a well-defined problem, making a simultaneous measurement of the ocean's temperature and salinity in remote sensing possible. Measuring temperature and salinity in unknown conditions with the findings presented in this work constitutes an important next step currently in planning.

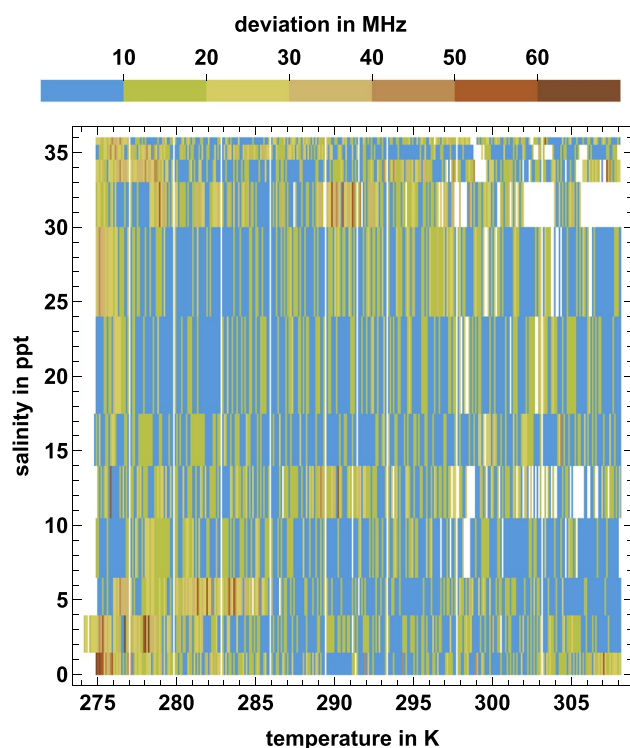


Fig. 5 Deviation of every data point from the polynomial shown in Figs. 3 and 4. If there was more than one point for a single temperature and salinity, these were averaged into a single point in this figure. Deviation is below 10MHz and 2% for over 50% of points and below 40MHz and 6% for more than 97% of points. White spaces indicate a lack of data at this point

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

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